

Problem 1.2**(a)**

$\sqrt{(-1)(-1)} = \sqrt{(-1)}\sqrt{(-1)}$ is wrong, because:

$$a \geq 0 \vee b \geq 0 \implies \sqrt{a \cdot b} = \sqrt{a} \cdot \sqrt{b}$$

(b)

Proof. Assume $1 = -1$ is true, we can add 3 to both sides of the equation yields:

$$4 = 2$$

Then we divide both sides by 2, we obtain:

$$2 = 1$$

which completes the proof. □

(c)

Proof. We want to show that

$$\sqrt{r \cdot s} = \sqrt{r} \cdot \sqrt{s}$$

Since r and s are positive real numbers as given, we can let $\sqrt{r} = a$ and $\sqrt{s} = b$, where a and b are positive real numbers. Squaring that we obtain $r = a^2$ and $s = b^2$, it follows that:

$$r \cdot s = a^2 \cdot b^2 = (a \cdot b)^2$$

Since a and b are positive real numbers, their product must also be a positive real number. Therefore, by the standard convention, $a \cdot b$ is the unique positive square root of $r \cdot s$, which means:

$$\sqrt{r \cdot s} = a \cdot b$$

Substituting $a = \sqrt{r}$ and $b = \sqrt{s}$ back into the equation, we obtain:

$$\sqrt{r \cdot s} = \sqrt{r} \cdot \sqrt{s}$$

which completes the proof. □